SCHOOL OF
PHYSICAL SCIENCES
AND NANOTECHNOLOGY

Scattering of a Klein-Gordon particle by an Exponential-Tanh Potential

Relativistic Quantum Mechanics Final Presentation

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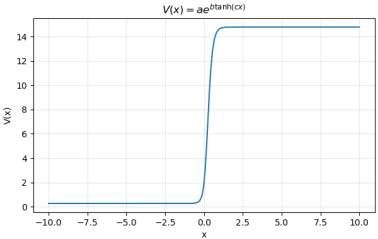

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Why Study Klein–Gordon and Superradiance?

- Use of KGE to study relativistic scattering.
- Classical systems: the reflected flux is *always smaller* than the incoming one ($R < 1$).
- Relativistic systems: reflected flux exceed the incoming one ($R > 1$), the wave is **amplified**.
- **Superradiance**, it corresponds to extracting energy from the system.
- Superradiance appears in rotating black holes possibly related to understanding dark matter.

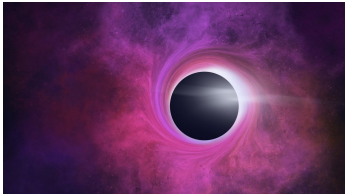


Figure: Rotating Black Hole

Concepts

What is Klein–Gordon Equation?

Study particles move close to the speed of light and the KGE is built to match relativity.

Key Concept

1. Klein Gordon Equation in 1D:

$$\frac{d^2\varphi}{dx^2} + \left[(E - V(x))^2 - m^2\right]\varphi = 0 \tag{1}$$

2. The energy enters **quadratically**, so new behaviors appear as superradiance

The Exponential–Tanh Potential

What does the potential look like?

Potential

$$V(x) = ae^{b\tanh(cx)} \tag{2}$$

Using the limits of the tanh function:

$$\tanh(cx) \longrightarrow -1 \quad \text{as} \quad x \to -\infty,$$
$$\tanh(cx) \longrightarrow +1 \quad \text{as} \quad x \to +\infty.$$

So the potential becomes:

$$V_L = ae^{-b}, \qquad V_R = ae^{b}.$$

Visual explanation

Key idea

The potential is smooth but asymmetric creating two distinct regions where the particle behaves differently.

$$E - V_R < 0$$

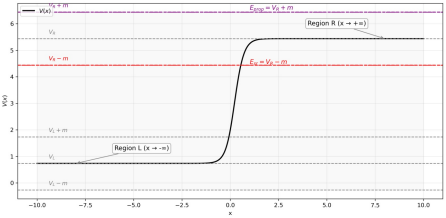


Figure: Asymptotic behavior of the potential.

Superradiance

The Superradiant Condition

The right region allows propagation

$$(E - V_R)^2 > m^2 \tag{3}$$

The transmitted flux becomes negative:

$$E - V_R < 0 \tag{4}$$

Together, the reflected wave is amplified

$$R > 1 \tag{5}$$

Riccati Log-Derivative Method

The key idea: Logarithmic Derivative

If we try to solve the Klein–Gordon equation by integrating the wavefunction $\varphi(x)$ directly, the solution can blow up.

$$y(x) = \frac{\varphi'(x)}{\varphi(x)} \tag{6}$$

Stable even if $\varphi(x)$ becomes large or small.

Riccati variable $y(x)$ removes the unstable exponential factors.

Numerical Method

Why Numerical Methods?

Goal: Extract R(E) and T(E) for the Klein–Gordon equation with strong numerical stability.
The problem: Analytical solutions for this potential are not available, so a robust numerical method is required.

Solution: Riccati Log-Derivative Method (RLM)

- Replaces $\varphi(x)$ with the smoother quantity

$$y(x) = \frac{\varphi'(x)}{\varphi(x)}.$$

- Removes dependence on the wavefunction normalization.
- Provides high numerical stability even in relativistic and superradiant regimes.

Riccati Equation

From Klein Gordon to Riccati Equation

$$\varphi''(x) + \left[(E - V(x))^2 - m^2\right]\varphi(x) = 0.$$

Define the local relativistic wave number

$$k^2(x) = (E - V(x))^2 - m^2.$$

Introduce the logarithmic derivative

$$y(x) = \frac{\varphi'(x)}{\varphi(x)}.$$

Differentiate $y(x)$, substitute the KG equation and get Riccati equation

$$y'(x) = \frac{\varphi''(x)}{\varphi(x)} - \left(\frac{\varphi'(x)}{\varphi(x)}\right)^2 = -k^2(x) - y^2(x) \rightarrow \boxed{y'(x) = -k^2(x) - y^2(x)}$$

Right Asymptotic Boundary Condition

The potential approaches a constant value,

$$V(x) \xrightarrow[x\to+\infty]{} V_R,$$

so the Klein–Gordon equation admits plane wave solutions. **Outgoing only physical condition:**

$$\varphi(x) \sim e^{ik_R x}, \qquad k_R = \sqrt{(E - V_R)^2 - m^2}.$$

Corresponding logarithmic derivative:

$$y(x_R) = \frac{\varphi'(x_R)}{\varphi(x_R)} = ik_R.$$

This value is used as the starting point for backward integration:

$$\boxed{y(x_R) = ik_R}$$

Numerical Backward Integration (RK4 Method)

Setup:

- Discretize the domain $[x_L, x_R]$.
- Start from the outgoing boundary condition:

$$y(x_R) = ik_R.$$

RK4 Integration (backwards):

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4),$$

with

$$k_i = f(x, y), \qquad f(x, y) = -k^2(x) - y^2.$$

Notes:

- Integrate from x_R to x_L.
- Cap $|y|$ to avoid overflow.

Output: The value $y(x_L)$ used to compute $R(E)$.

Reflection and Transmission Coefficients

Left asymptotic form:

$$\varphi(x) = e^{ik_L x} + R\, e^{-ik_L x}, \qquad k_L = \sqrt{(E - V_L)^2 - m^2}.$$

Logarithmic derivative at x_L:

$$y(x_L) = \frac{\varphi'(x_L)}{\varphi(x_L)}.$$

Reflection amplitude:

$$\boxed{R = e^{2ik_L x_L}\, \frac{ik_L - y(x_L)}{ik_L + y(x_L)}}$$

Reflection coefficient:

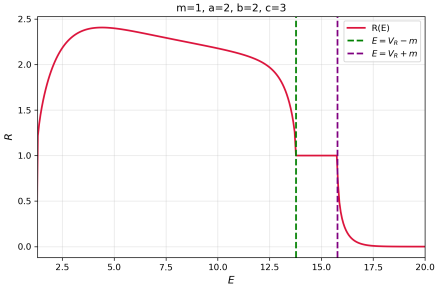
$$\boxed{R(E) = |R|^2}$$

Transmission coefficient (flux conservation):

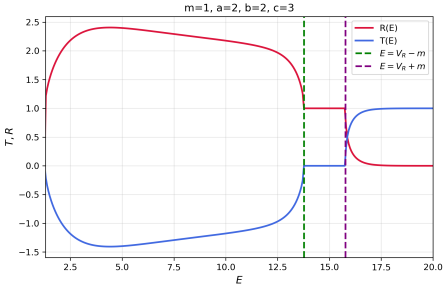
$$\boxed{T(E) = 1 - R(E)}$$

Results

Superradiance–Case 1



$R(E)$

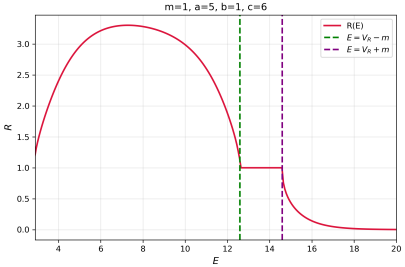


$R(E)$ and $T(E)$

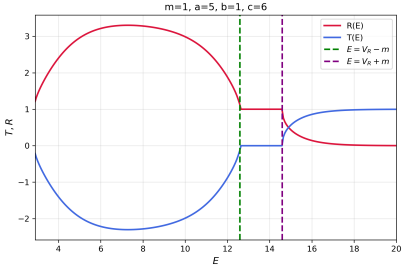
Main features:

- For $E < V_R - m$ there is a **superradiant window** with $R(E) > 1$ and $T(E) < 0$.
- At $E = V_R - m$ the transmitted mode changes from evanescent to propagating and $T(E)$ jumps to zero.
- For $E > V_R + m$, both sides are propagating: $R(E)$ decreases towards 0 and $T(E)$ increases towards 1.

Superradiance–Case 2



$R(E)$



$R(E)$ and $T(E)$

Main features:

- The **superradiant window** persists, with $R(E) > 1$ and $T(E) < 0$, but its width and peak value change compared to Case 1.
- The thresholds $E = V_R - m$ and $E = V_R + m$ shift, modifying the onset and end of superradiance.
- At high energies, the potential becomes effectively transparent: $R(E) \to 0$ and $T(E) \to 1$.

Interactive Manipulate

Interactive Manipulate

Figure: SCAN ME! (Own creation)

Discussion

Understanding the Scattering Behavior

Overview

Regimes:

Superradiance $R > 1$

Evanescent plateau $R = 1, T = 0$

High energy transparency $R = 0, T = 1$

The Role of Asymptotic Region

Superradiance requires:

$$(E - V_R)^2 > m^2, \qquad E - V_R < 0.$$

Threshold:

$$E < V_R - m.$$

The parameter V_R shifts the entire superradiant window.
Higher values of V_R imply that superradiance occurs at higher energies.

Conclusions

Conclusions

- The Riccati Log–Derivative Method proved to be a **stable and efficient** numerical approach for solving relativistic scattering in the Klein–Gordon equation.
- The method successfully identified the **superradiant regime**, characterized by enhanced reflection $R(E) > 1$ and negative transmission $T(E)$.
- The location and width of the superradiant window depend sensitively on the potential parameters, illustrating the method's ability to resolve fine physical structure.
- At high energies, both parameter sets converge to the expected relativistic transparency limit ($R(E) \to 0$, $T(E) \to 1$).
- **Future work:** Extend the analysis to more general potentials, compare with WKB, complex scaling methods or other numerical methods.

Thank you!

Any questions?

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